

Unrealized Capital Gains Tax, Business Investment, and Wealth Inequality*

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February 2026

Abstract

Unrealized capital gains constitute a large share of wealth at the top of the U.S. wealth distribution and are concentrated in privately held businesses, yet they remain untaxed under the current realization-based capital-gains tax system. This paper studies the welfare consequences of replacing realization-based taxation with an accrual-based (mark-to-market) tax regime, with particular emphasis on entrepreneurial business assets. Using a quantitative general-equilibrium model, I find that an economy-wide switch to accrual-based taxation reduces private business production by 1.5% and aggregate output by 0.4%, because entry deterrence outweighs the efficiency gains from mitigating capital lock-in. In contrast, applying accrual taxation only to wealthy households yields a consumption-equivalent welfare gain of 0.2%. By targeting wealthy households that are more likely to be incumbent entrepreneurs, this reform dampens entry deterrence while still reducing lock-in; as a result, entrepreneurial production increases by 0.4% and aggregate output rises by 0.2%.

*I am indebted to my advisers Fabio Ghironi, Brian Greaney, and Ellen McGrattan for their invaluable guidance and support. I also thank Marco Bassetto, Philip Bond, V. V. Chari, Yu-chin Chen, Mariacristina De Nardi, Jonathan Dingel, Stephie Fried, Ippei Fujiwara, Kyle Herkenhoff, Yuhei Miyauchi, Toshihiko Mukoyama, Andrii Parkhomenko, Fernando Parro, John Stachurski, and Stephen Turnovsky for their comments and suggestions.

1 Introduction

In the United States, unrealized (accrued) capital gains—the increase in asset values that has not yet been realized through sale—account for roughly 42% of the net worth held by households in the top 1% of the wealth distribution, and most of these gains reflect appreciation in *privately held businesses*. Because such gains are untaxed under the current realization-based system, policymakers have recently debated broadening the capital-gains tax base to include unrealized gains, both at the federal level and in several U.S. states.¹

Motivated by these policy discussions and by the empirical importance of private-business capital gains at the top of the distribution, this paper studies how accrual-based capital-gains taxation affects welfare, with particular emphasis on entrepreneurial (privately held business) assets. Existing studies such as [Bhandari et al. \(2024\)](#) and [Azevedo et al. \(2025\)](#) examine the implications of taxing business capital values, but they primarily focus on firm dynamics; the aggregate welfare consequences remain comparatively less explored. This paper fills that gap by quantifying the welfare effects of accrual-based taxation in a quantitative general-equilibrium model calibrated to match salient features of the U.S. wealth distribution. To clarify the underlying economic forces, I also develop a stylized model that isolates two central mechanisms—*capital lock-in* and *firm entry deterrence*—and uses them to interpret the quantitative results.

The *capital lock-in* effect is a classic implication of realization-based capital-gains taxation (e.g., [Auerbach \(1991\)](#)). Because tax liability is triggered only upon sale, entrepreneurs have an incentive to defer realization and retain existing firms even when reallocating effort toward new, more productive business opportunities would be efficient. By continuing to operate an appreciated business, they postpone tax payments and allow unrealized gains to compound tax free; by contrast, selling triggers an immediate tax bill, and restarting exposes subsequent gains to repeated taxation.

Accrual-based taxation can, in principle, mitigate lock-in by taxing gains regardless of whether they are realized. At the same time, however, accrual taxation imposes tax liability precisely upon successful business creation—a margin that is untaxed under realization-based regimes. As a result,

¹The Biden administration has put forward a proposal to tax unrealized capital gains, and five U.S. states have introduced related legislation: AB 2088, the *Wealth Tax Act* (2019–20) in California; S 1570/A 3252, the *Billionaire Mark-to-Market Tax Act* (2023–24) in New York; HB 3039, the *Extremely High-Wealth Mark-to-Market Tax Act* (103rd GA, 2023–24) in Illinois; HB 1473, the *Washington Extreme-Wealth Tax* (2023–24) in Washington; and H 827 (2023–24), an act applying the personal income tax to unrealized gains in Vermont.

it may *deter firm entry* by reducing the expected payoff to starting a business. The central question, therefore, is which force dominates: does accrual taxation improve allocative efficiency primarily by reducing lock-in, or does it instead suppress entrepreneurial activity by discouraging entry?

I address this question in a quantitative general-equilibrium model in which a corporate sector produces final goods using labor, capital, and differentiated intermediate inputs supplied by a private business sector. Households are heterogeneous and choose between working and entrepreneurship. Entrepreneurs operate monopolistically competitive private businesses and can either continue to manage their firms or transfer them through sale. This environment provides a unified and empirically disciplined framework for evaluating how alternative capital-gains tax regimes shape entrepreneurial activity, aggregate production, and welfare.

To organize the analysis, I first develop a tractable stylized version of the quantitative model and compare the decentralized equilibrium to a constrained social planner (who cannot use lump-sum transfers). Even absent capital-gains taxation, the model economy features wedges that distort both the intensive margin (capital investment) and the extensive margin (occupational choice, including firm entry and the decision to sell). On the intensive margin, monopolistic competition generates markups that depress private incentives to accumulate capital, leading to underinvestment relative to the planner. On the extensive margin, the decentralized equilibrium exhibits inefficiently low firm entry and too few business transfers. Entry is inefficiently low because market power reduces the private return to creating new firms. Business transfers are inefficiently rare because individual entrepreneurs do not internalize the aggregate benefit of higher turnover: selling allows entrepreneurs to restart more quickly and expands the variety of intermediate inputs, generating an aggregate-demand (love-of-variety) externality. A realization-based capital-gains tax amplifies this distortion by further discouraging sales and strengthening lock-in. Accrual-based taxation, in contrast, mitigates lock-in but exacerbates entry deterrence by taxing successful creation.

These mechanisms carry over to the quantitative model. Replacing the current realization-based regime with accrual-based taxation reduces lock-in but strengthens entry deterrence. In the calibrated economy, the entry-deterrence effect dominates: private-business production falls by 1.5%, and final-good output falls by 0.4%. The decline in final-good output is smaller than the decline in private-business production because discouraged potential entrants become workers, increasing labor supply.

I then evaluate a hybrid regime that resembles prominent policy proposals: accrual-based taxation applies only to wealthy households, while the rest of the population remains subject to realization-based taxation.² This targeted design can dampen the entry-deterrence margin because top-wealth households are more likely to be incumbent entrepreneurs than marginal entrants, while still reducing lock-in among high-wealth business owners. Under a policy that applies accrual-based taxation to the top 0.5% of wealthy households, entrepreneurial production rises by 0.4%, aggregate output increases by 0.2%, and consumption-equivalent welfare increases by 0.2% relative to the realization-based status quo.

However, these results are highly sensitive to the choice of the wealth cutoff. On the one hand, targeting only very wealthy households dampens firm-entry deterrence, but applying accrual-based taxation to too small a population generates only limited efficiency gains from reducing lock-in. On the other hand, broadening the tax base too much strengthens the entry-deterrence effect and can offset the gains from reduced lock-in. These findings suggest that designing a hybrid accrual-based tax that generates welfare gains requires careful targeting.

Finally, the analysis focuses on long-run stationary effects. It therefore does not capture the potentially large transitional revenues that could arise from taxing unrealized gains already accumulated by wealthy households.³ In the long run, however, accrual-based taxation has only a limited effect on tax revenue because it mainly changes the timing of capital-gains taxation, especially when capital losses are fully refunded.

1.1 Related Literature

This paper contributes to three strands of literature. First, it relates to the literature on capital-gains taxation, capital lock-in, and business transfers. A classic implication of realization-based capital-gains taxation is that it discourages asset sales by allowing taxpayers to defer tax liabilities until realization. [Chari et al. \(2005\)](#) study this lock-in effect in the context of entrepreneurial business transfers. I build on this insight by incorporating accrual-based taxation and showing that it creates a central trade-off for entrepreneurship. On the one hand, accrual-based taxation

²The Biden administration’s proposal applies accrual-based taxation to households with wealth above \$100 million, which corresponds to approximately the top 0.02% of the wealth distribution.

³The top 0.02% of wealthy households hold about \$6 trillion in unrealized capital gains, and about 64% of these gains are held in private-business shares [Saez et al. \(2021\)](#).

mitigates lock-in by taxing gains regardless of whether the business is sold. On the other hand, it may deter firm entry by reducing the after-tax payoff from successful business creation. This trade-off is closely related to [Bhandari et al. \(2024\)](#), who quantitatively study taxes on business capital values. My model also incorporates an accrual-tax feature that rebates previously paid taxes when capital losses occur, allowing the analysis to capture the insurance channel emphasized by [Azevedo et al. \(2025\)](#). Relative to this firm-dynamics literature, my contribution is to quantify the aggregate welfare consequences of accrual-based capital-gains taxation in a model disciplined by key features of the U.S. wealth distribution. The framework is also suited to evaluating hybrid regimes in which accrual-based taxation applies only to very wealthy households, while the rest of the population remains subject to realization-based taxation.

Second, this paper contributes to the wealth-inequality literature by emphasizing the role of unrealized capital gains embedded in privately held businesses. A large body of work studies entrepreneurship and wealth concentration, including [Quadrini \(2000\)](#), [Cagetti and De Nardi \(2006\)](#), and [Gomez and Gouin-Bonenfant \(2024\)](#). Another strand studies heterogeneous returns to capital as an important source of wealth inequality, including [Benhabib et al. \(2011\)](#), [Bach et al. \(2020\)](#), [Fagereng et al. \(2020\)](#), [Xavier \(2021\)](#), and [Brüggemann and Mahone \(2025\)](#). Related work also highlights the role of illiquid, high-return assets in wealth accumulation, such as [Robbins \(2018\)](#); [Fagereng et al. \(2019, 2024\)](#); [Smith et al. \(2023\)](#). My paper complements these studies by focusing on privately held business wealth as an illiquid entrepreneurial asset whose sale decision determines the timing of capital-gains taxation. This allows me to study how the tax treatment of unrealized business gains affects both wealth concentration and business reallocation.

Third, this paper contributes to the literature on taxation and entrepreneurship. Existing work studies how tax policy affects entrepreneurial decisions, wealth accumulation, and aggregate outcomes, including [Kitao \(2008\)](#), [Cagetti and De Nardi \(2009\)](#), [Scheuer \(2014\)](#), [Brüggemann \(2021\)](#), [Guvenen et al. \(2023\)](#), and [Boar and Midrigan \(2023\)](#). I add to this literature by introducing endogenous business transfers as a key margin of adjustment. This margin is important because capital-gains taxation affects not only the decision to become an entrepreneur, but also the decision to retain or sell an existing business. The model therefore captures how alternative capital-gains tax regimes shape both firm entry and capital lock-in, and how the interaction of these two margins determines aggregate production and welfare.

The remainder of the paper is organized as follows. Section 2 presents the stylized model and clarifies the two core mechanisms. Section 3 sets out the quantitative model under the current realization-based tax system. Section 4 describes the calibration. Section 5 extends the model to incorporate accrual-based taxation. Section 6 reports the counterfactual results, and Section 7 concludes.

2 Simple model

This section develops a stylized model to clarify the mechanisms through which capital-gains tax reforms affect entrepreneurial activity. The framework highlights responses on both the intensive margin (capital used by operating private businesses) and the extensive margin (occupational choice and business transfer). It combines (i) the business-transfer mechanism and occupational-choice structure in [Chari et al. \(2005\)](#), (ii) differentiated intermediate inputs produced by private businesses as in [Guvenen et al. \(2023\)](#) and [Rotberg and Steinberg \(2024\)](#), and (iii) a private-business technology in which capital is the sole input, following [Cagetti and De Nardi \(2006\)](#). I first characterize the baseline economy without capital-gains taxation and then study reforms under realization-based and accrual-based capital-gains taxes.

2.1 Environment

[Figure 1](#) illustrates the model structure. The economy is populated by a unit mass of infinitely lived households and a competitive “bank” sector (a representative price-taking firm) with two divisions: (i) final-good production and (ii) mergers & acquisitions (M&A). Households are heterogeneous and choose an occupation: they either supply labor as workers or operate private businesses as entrepreneurs. The final-good division purchases differentiated intermediate inputs from operating private businesses. The M&A division acquires businesses from entrepreneurs and operates them thereafter, supplying intermediate inputs as well.

Capital used in private-business production is rented in a global market at an exogenously given interest rate r . Accordingly, this section takes r as given and focuses on a partial-equilibrium environment. In a later quantitative section, I endogenize the capital market and study general-equilibrium implications.

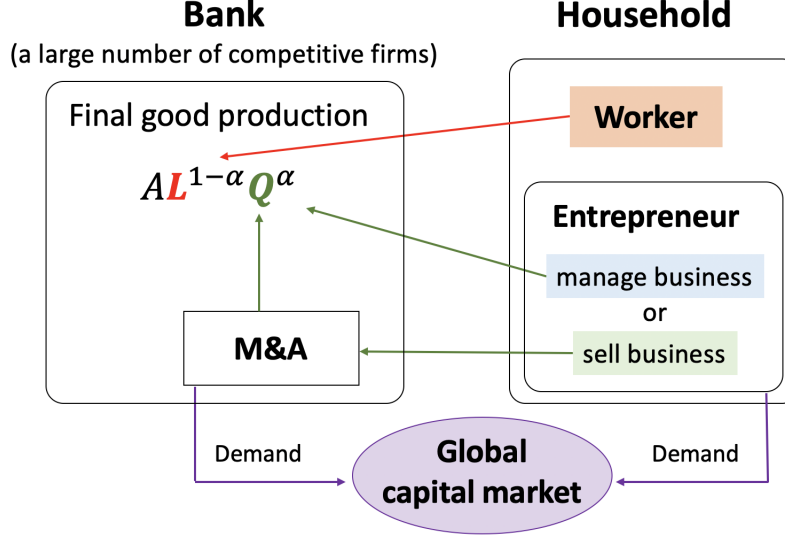


Figure 1: Production structure in the simple model economy

Households. Households are heterogeneous in business-idea ability $\theta \in [0, 1]$, which governs the probability of generating a viable startup opportunity. Ability is time-invariant and distributed according to a probability measure H with density h . Households cannot save (there is no storage technology), so they consume current income each period. Preferences are linear:

$$u(c_t) = c_t.$$

For convenience, I adopt the stationary representative-agent benchmark normalization,

$$r = \frac{1 - \beta}{\beta}.$$

At the beginning of life, each household makes a lifetime occupational choice among: (i) worker; (ii) entrepreneur who starts a business and then sells it (*start-and-sell*); or (iii) entrepreneur who starts a business and then manages it (*start-and-manage*). The household's problem is to choose the occupation that maximizes its lifetime value,

$$\max \{V^w, V^s(z = 0, \theta), V^m(z = 0, \theta)\}. \quad (1)$$

Workers. V^w denotes the value function of a worker. A worker earns wage income w each period and consumes it. The value function satisfies

$$V^w = w + \beta V^w \quad \Rightarrow \quad V^w = \frac{w}{1 - \beta}. \quad (2)$$

Entrepreneurs who start and sell. $V^s(z, \theta)$ denotes the value function of the *start-and-sell* entrepreneur. If a household chooses this occupation, he devotes one period to startup business. In the following period, he becomes a business owner with probability θ . Business quality is $z \in [0, \bar{z}]$, where $z = 0$ denotes the non-ownership state and $\bar{z}$ denotes the quality of an operating business. For simplicity, here I assume a single quality level. Upon successful creation, the household draws quality $\bar{z}$, sells the business at price $p(\bar{z})$, and returns to the non-ownership state in the subsequent period. The value functions satisfy

$$\begin{cases} V^s(0, \theta) = \beta [(1 - \theta)V^s(0, \theta) + \theta V^s(\bar{z}, \theta)], \\ V^s(\bar{z}, \theta) = p(\bar{z}) + \beta V^s(0, \theta). \end{cases} \quad (3)$$

Entrepreneurs who start and manage. $V^m(z, \theta)$ denotes the value function of the *start-and-manage* entrepreneur. The startup stage is identical to that of the start-and-sell entrepreneur. Conditional on success, the household continues to operate the business and earns operating profit. Business output is linear in capital:

$$x = \bar{z}k,$$

where k is rented at rate r . A managed business survives into the next period with probability $\nu \in (0, 1)$. The value functions satisfy

$$\begin{cases} V^m(0, \theta) = \beta [(1 - \theta)V^m(0, \theta) + \theta V^m(\bar{z}, \theta)], \\ V^m(\bar{z}, \theta) = \max_{k \geq 0} \pi_m(k) + \beta [(1 - \nu)V^m(0, \theta) + \nu V^m(\bar{z}, \theta)], \end{cases} \quad (4)$$

where operating profit is

$$\pi_m(k) = \xi(x)x - rk \quad (5)$$

and $\xi(\cdot)$ denotes the inverse demand schedule for a differentiated intermediate input (derived below from final-good production). Let $\pi_m^* \equiv \max_{k \geq 0} \pi_m(k)$ denote optimal operating profit.

In the quantitative model, households are additionally heterogeneous in labor productivity, multiple business-quality states are allowed, and occupational choice is made each period rather than only once at the beginning of life.

Bank. The competitive bank sector has two divisions: final-good production and M&A. The final-good division produces the final good using labor L and a CES aggregate Q of differentiated intermediate inputs:

$$Y = AL^{1-\alpha}Q^\alpha, \quad \alpha \in (0, 1). \quad (6)$$

The intermediate input is

$$Q = \left(\int x_i^\mu di \right)^{1/\mu}, \quad \mu \in (0, 1), \quad (7)$$

where i indexes operating businesses/varieties. This aggregator implies an elasticity of substitution $1/(1 - \mu) > 1$ across varieties.

The M&A division acquires businesses from entrepreneurs but operates them less efficiently. Let $\lambda \in (0, 1)$ denote the efficiency wedge (equivalently, entrepreneurs may derive non-pecuniary private benefits from control), following [Chari et al. \(2005\)](#). If an acquired business is operated by M&A, output is

$$x = \lambda \bar{z}k,$$

with capital rented at rate r .

Given the CES structure, the (price-taking) final-good division's demand for any variety i implies the inverse demand schedule

$$\xi(x_i) = \alpha AL^{1-\alpha}Q^{\alpha-\mu}x_i^{\mu-1}. \quad (8)$$

Accordingly, an entrepreneur-managed business chooses k to maximize (5), and an M&A-operated business chooses k to maximize

$$\pi_b(k) = \xi(x)x - rk, \quad \text{with } x = \lambda \bar{z}k, \quad (9)$$

with optimal profit $\pi_b^* \equiv \max_{k \geq 0} \pi_b(k)$.

Because the M&A division is competitive, the purchase price $p(\bar{z})$ equals the present value of operating profits while the business survives

$$p(\bar{z}) = \pi_b^* + \beta\nu p(\bar{z}) \quad \Longleftrightarrow \quad p(\bar{z}) = \frac{\pi_b^*}{1 - \beta\nu}. \quad (10)$$

2.2 Equilibrium without capital gains tax

This subsection characterizes occupational choice in the baseline economy, taking (w, r) and the induced objects $(p(\bar{z}), \pi_m^*)$ as given.

Occupational cutoffs. Cutoff values of θ are pinned down by comparing value functions evaluated at $z = 0$.

First, in a two-way comparison between worker and start-and-sell, the indifference cutoff is

$$\theta^{sw} = \frac{w}{\beta(p(\bar{z}) - w)}. \quad (11)$$

Second, in a two-way comparison between worker and start-and-manage, the indifference cutoff is

$$\theta^{mw} = \frac{(1 - \beta\nu) w}{\beta(\pi_m^* - w)}. \quad (12)$$

Third, the cutoff between start-and-sell and start-and-manage is

$$\theta^{sm} = \frac{(1 - \beta\nu) p(\bar{z}) - \pi_m^*}{\beta(\pi_m^* - p(\bar{z}))}. \quad (13)$$

When all three occupations are active, the equilibrium mapping is

$$\theta < \theta^{mw} \Rightarrow \text{worker}, \quad \theta^{mw} \leq \theta < \theta^{sm} \Rightarrow \text{start-and-manage}, \quad \theta^{sm} \leq \theta \Rightarrow \text{start-and-sell}.$$

Using (10) and the homogeneity of profits under (8), one can express θ^{sm} as a function of the

efficiency wedge λ . In particular, $\pi_b^*/\pi_m^* = \lambda^{\frac{\mu}{1-\mu}}$, and substituting into (13) yields

$$\theta^{sm} = \frac{(1 - \beta\nu) \left(\lambda^{\frac{\mu}{1-\mu}} - 1 \right)}{\beta \left(1 - \beta\nu - \lambda^{\frac{\mu}{1-\mu}} \right)}. \quad (14)$$

I focus on the empirically relevant case in which all three occupations are simultaneously active. A convenient set of sufficient conditions is as follows.

Proposition 1. If $\frac{(1+\beta)(1-\beta\nu)}{1-\beta\nu+\beta} \leq \lambda^{\frac{\mu}{1-\mu}} \leq 1 - \frac{\beta\nu w}{\pi_m^*}$, then all three occupations are active. Households with $\theta < \theta^{mw}$ become workers; those with $\theta^{mw} \leq \theta < \theta^{sm}$ become start-and-manage entrepreneurs; and those with $\theta \geq \theta^{sm}$ become start-and-sell entrepreneurs.

Figure 2 illustrates the value functions and the associated cutoff levels.

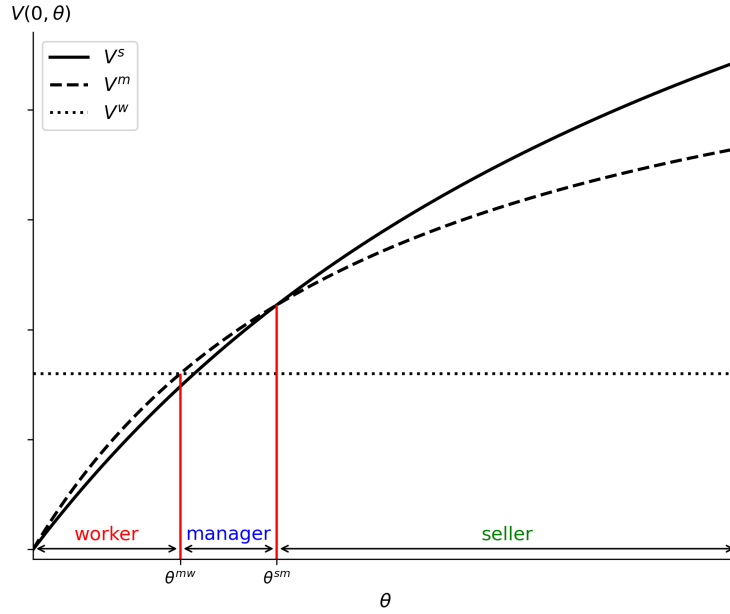


Figure 2: Occupational choice cutoff levels in the baseline economy

Stationary measures and labor supply. Let f_m denote the stationary measure of *operating* entrepreneur-managed businesses. For a type- θ start-and-manage entrepreneur, the stationary probability of being in the operating state is $\theta/(1 - \nu + \theta)$. Hence

$$f_m = \int_{\theta^{mw}}^{\theta^{sm}} \frac{\theta}{1 - \nu + \theta} h(\theta) d\theta. \quad (15)$$

Let f_b denote the stationary measure of businesses operated by the bank's M&A division. For a type- θ start-and-sell entrepreneur, the stationary probability of being in the (pre-sale) ownership state is $\theta/(1 + \theta)$, so the flow of sales to the bank is $\theta/(1 + \theta)$ per period. Since bank-operated businesses survive with probability ν , the stationary stock equals the flow divided by the exit rate $1 - \nu$:

$$f_b = \frac{1}{1 - \nu} \int_{\theta^{sm}}^1 \frac{\theta}{1 + \theta} h(\theta) d\theta. \quad (16)$$

Because only workers supply labor to final-good production, aggregate labor input is

$$L = H(\theta^{mw}). \quad (17)$$

2.3 Introduction of a realization-based capital-gains tax

Suppose realized capital gains are taxed at rate τ . Under realization-based taxation, only start-and-sell entrepreneurs pay the tax, upon sale. The start-and-sell value function becomes

$$\begin{cases} V^s(0, \theta) = \beta [(1 - \theta)V^s(0, \theta) + \theta V^s(\bar{z}, \theta)], \\ V^s(\bar{z}, \theta) = (1 - \tau)p(\bar{z}) + \beta V^s(0, \theta). \end{cases} \quad (18)$$

Thus the effective sale price is reduced to $(1 - \tau)p(\bar{z})$, and the seller-manager cutoff shifts to

$$\theta_{\text{real}}^{sm} = \frac{(1 - \beta\nu)(1 - \tau)p(\bar{z}) - \pi_m^*}{\beta(\pi_m^* - (1 - \tau)p(\bar{z}))}. \quad (19)$$

Proposition 2. If τ is sufficiently small so that all three occupations remain active, then

$$\theta_{\text{real}}^{sm} > \theta^{sm}.$$

A realization-based capital-gains tax therefore raises the threshold for selecting the start-and-sell entrepreneur: the measure of sellers falls while the measure of managers rises. Intuitively, some high-ability households (high θ) that would have sold their businesses in the no-tax economy instead choose to retain and operate them when realizations are taxed. Following [Chari et al. \(2005\)](#), I refer to this reallocation as a *capital lock-in* effect. Figure 3 illustrates this mechanism.

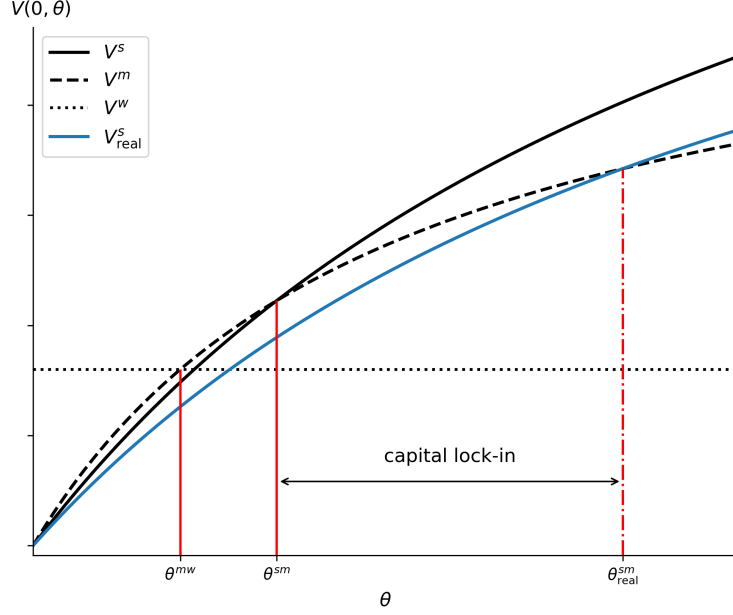


Figure 3: Occupational cutoffs under a realization-based capital-gains tax

2.4 Introduction of an accrual-based capital-gains tax

Suppose instead that capital gains are taxed upon accrual at rate τ . In this stylized environment, business quality is constant while operating, so accrual taxation is equivalent to taxing the business value $p(\bar{z})$ when the household successfully creates a business (i.e., when it transitions from $z = 0$ to $z = \bar{z}$). If the business subsequently exits (a capital-loss event), previously paid taxes are rebated. Many policy proposals implement loss offsets through carry-forward provisions; for tractability, I abstract from carry-forward constraints and assume that losses are fully rebated contemporaneously.

Under accrual taxation, both the seller's and manager's value functions are modified.

Seller.

$$\begin{cases} V^s(0, \theta) = \beta \left[(1 - \theta)V^s(0, \theta) + \theta \left(V^s(\bar{z}, \theta) - \tau p(\bar{z}) \right) \right], \\ V^s(\bar{z}, \theta) = p(\bar{z}) + \beta V^s(0, \theta). \end{cases} \quad (20)$$

Manager.

$$\begin{cases} V^m(0, \theta) = \beta \left[(1 - \theta)V^m(0, \theta) + \theta \left(V^m(\bar{z}, \theta) - \tau p(\bar{z}) \right) \right], \\ V^m(\bar{z}, \theta) = \pi_m^* + \beta \left[(1 - \nu) \left(V^m(0, \theta) + \tau p(\bar{z}) \right) + \nu V^m(\bar{z}, \theta) \right]. \end{cases} \quad (21)$$

Because the tax is levied upon accrual, households pay $\tau p(\bar{z})$ upon successful business creation. If the business exits, the associated “capital loss” triggers a rebate of the same amount under the full-refund assumption.

These modifications shift the occupational cutoffs. The worker–manager cutoff becomes

$$\theta_{\text{accrual}}^{mw} = \frac{w(1 - \beta\nu)}{\beta(\pi_m^* - w - (1 - \beta)\tau p(\bar{z}))}, \quad (22)$$

and the seller–manager cutoff becomes

$$\theta_{\text{accrual}}^{sm} = \frac{(\pi_m^* + \beta\tau p(\bar{z})) - (1 - \beta\nu(1 - \tau))p(\bar{z})}{\beta[p(\bar{z}) - (\pi_m^* + \beta\tau p(\bar{z}))]}. \quad (23)$$

Proposition 3. If τ is sufficiently small that all three occupations remain active, then

$$\theta_{\text{real}}^{sm} > \theta_{\text{accrual}}^{sm} > \theta^{sm} \quad \text{and} \quad \theta_{\text{accrual}}^{mw} > \theta^{mw} = \theta_{\text{real}}^{mw}.$$

Thus, relative to realization-based taxation, accrual taxation mitigates capital lock-in (by lowering the cutoff for selling relative to $\theta_{\text{real}}^{sm}$), but it discourages entrepreneurial entry by raising the worker–manager cutoff.

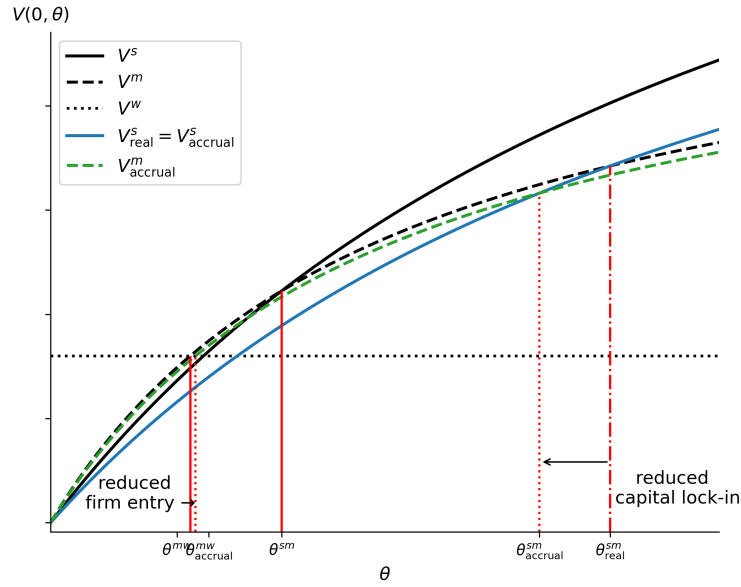


Figure 4: Occupational cutoffs under an accrual-based capital-gains tax

2.5 Social planner

This subsection studies a constrained social planner's problem to characterize distortions in the model and to assess whether capital lock-in is socially distortionary. To focus on allocative inefficiencies rather than redistribution, I assume the planner cannot use lump-sum taxes or transfers.⁴

With linear utility and a unit mass of households, the planner's objective is to maximize discounted aggregate consumption:

$$\max \sum_{t=0}^{\infty} \beta^t \int u(c_t) dH(\theta) = \sum_{t=0}^{\infty} \beta^t C_t, \quad (24)$$

The resource constraint is

$$Y_t - rK_t = C_t, \quad (25)$$

where K_t is aggregate capital rented for private-business production. Because capital is supplied by the global market, capital income rK_t accrues to foreigners and is subtracted from domestic resources.

Substituting the resource constraint into the objective yields the planner's problem

$$\max_{\theta^{mw}, \theta^{sm}, k_m, k_b} \sum_{t=0}^{\infty} \beta^t \{AL^{1-\alpha}Q^\alpha - r[f_mk_m + f_bk_b]\}. \quad (26)$$

where (L, Q, f_m, f_b) are determined by the occupational cutoffs and the induced stationary measures.

The key wedges in this environment are (i) a monopolistic-competition distortion in intermediate-good production and (ii) an externality through the dependence of aggregate demand on the mass of varieties (which affects the private return to operating and to entry). To clarify their implications, I compare the planner's allocation to the baseline competitive equilibrium along both the intensive and extensive margins.

⁴Equivalently, the planner is restricted to implementable allocations without non-distortionary fiscal instruments.

Intensive margin. Let k_m denote capital used by an entrepreneur-managed operating business and k_b denote capital used by a bank-operated business. Under (8), the planner chooses

$$k_m^{SP} = \left[\frac{\alpha A L^{1-\alpha} Q^{\alpha-\mu} \bar{z}^\mu}{r} \right]^{\frac{1}{1-\mu}} > k_m = \left[\frac{\mu \alpha A L^{1-\alpha} Q^{\alpha-\mu} \bar{z}^\mu}{r} \right]^{\frac{1}{1-\mu}}, \quad (27)$$

$$k_b^{SP} = \left[\frac{\alpha A L^{1-\alpha} Q^{\alpha-\mu} (\lambda \bar{z})^\mu}{r} \right]^{\frac{1}{1-\mu}} > k_b = \left[\frac{\mu \alpha A L^{1-\alpha} Q^{\alpha-\mu} (\lambda \bar{z})^\mu}{r} \right]^{\frac{1}{1-\mu}}. \quad (28)$$

Thus, the competitive equilibrium underinvests in private business capital because individual producers internalize only a fraction μ of the marginal social return—a standard markup distortion under monopolistic competition.

Extensive margin. The extensive margin is summarized by the occupational cutoffs θ^{mw} and θ^{sm} , which govern entry into entrepreneurship and the choice between managing and selling. In the equilibrium,

$$\theta_{SP}^{mw} < \theta^{mw},$$

implying that entrepreneurial entry is inefficiently low in the competitive equilibrium. Intuitively, market power in intermediate-good production depresses the private return to entry relative to the social return.

Moreover, I obtain

$$\theta_{SP}^{sm} < \theta^{sm},$$

so that the planner assigns a larger measure of entrepreneurs to the start-and-sell occupation. This difference reflects a trade-off between (i) the turnover benefit of selling—which increases the rate at which entrepreneurs can initiate new businesses—and (ii) the production inefficiency after acquisition by the bank, captured by λ . Formally, the sign of $\frac{\partial Q}{\partial \theta^{sm}}$ depends on

$$\underbrace{\frac{1}{1 - \nu + \theta^{sm}}}_{\text{marginal manager contribution}} - \underbrace{\frac{\lambda^{\frac{\mu}{1-\mu}}}{(1 - \nu)(1 + \theta^{sm})}}_{\text{marginal seller contribution}}. \quad (29)$$

This expression highlights the underlying trade-off: raising θ^{sm} increases the mass of managers and reduces the mass of sellers, and the net effect on Q depends on their relative contributions.

Managers and sellers differ along two dimensions. First, the turnover friction differs: the relevant denominator for managers is larger by $\nu\theta^{sm}$, reflecting that managers can start a new business only after exit, whereas sellers can restart immediately after selling. This makes the start-and-sell occupation relatively more effective at generating new businesses. Second, sellers' businesses are less productive after acquisition by the bank, reflected in the multiplicative penalty $\lambda^{\frac{\mu}{1-\mu}}$ in the seller term.

In the competitive equilibrium, individual households do not internalize the aggregate benefit of higher turnover from selling (an aggregate-demand externality), and they are discouraged from selling by the post-acquisition inefficiency in bank-operated production. Together, these forces raise θ^{sm} relative to the planner's choice, generating too few sellers. In the social planner's optimum, θ^{sm} is chosen to internalize these forces and to exactly balance this trade-off.

2.6 Comparison between the social planner and tax regimes

I now compare the allocations under capital-gains taxation to the constrained social planner's allocation. Along the intensive margin, the optimal capital choice under both tax regimes coincides with the baseline competitive equilibrium; hence, in both cases private business investment remains inefficiently low relative to the social planner. Capital-gains taxation, however, affects the extensive margin through occupational choice.

Corollary. If τ is sufficiently small that all three occupations remain active, then

$$\theta_{\text{real}}^{sm} > \theta_{\text{accrual}}^{sm} > \theta^{sm} > \theta_{SP}^{sm} \quad \text{and} \quad \theta_{\text{accrual}}^{mw} > \theta^{mw} = \theta_{\text{real}}^{mw} > \theta_{SP}^{mw}.$$

Under realization-based taxation, the capital lock-in mechanism in Proposition 2 reduces the measure of sellers. Comparing to the social planner makes clear that this reallocation is distortionary: the competitive equilibrium already features too few sellers relative to the planner. Switching from realization-based to accrual-based taxation mitigates this distortion, as implied by $\theta_{\text{real}}^{sm} > \theta_{\text{accrual}}^{sm}$. However, accrual-based taxation raises the entry cutoff θ^{mw} , discouraging entrepreneurial entry relative to realization-based taxation; relative to the planner, this entry deterrence is also distortionary. The key quantitative question is therefore whether the welfare gains

from mitigating capital lock-in outweigh the welfare losses from reduced firm entry. The next section addresses this question quantitatively.

3 Quantitative model

The quantitative model extends the simple model to match key empirical moments for aggregate production, the occupational distribution, entrepreneurial dynamics, and the wealth distribution.

Figure 5 provides an overview of the structure of the enriched model economy.

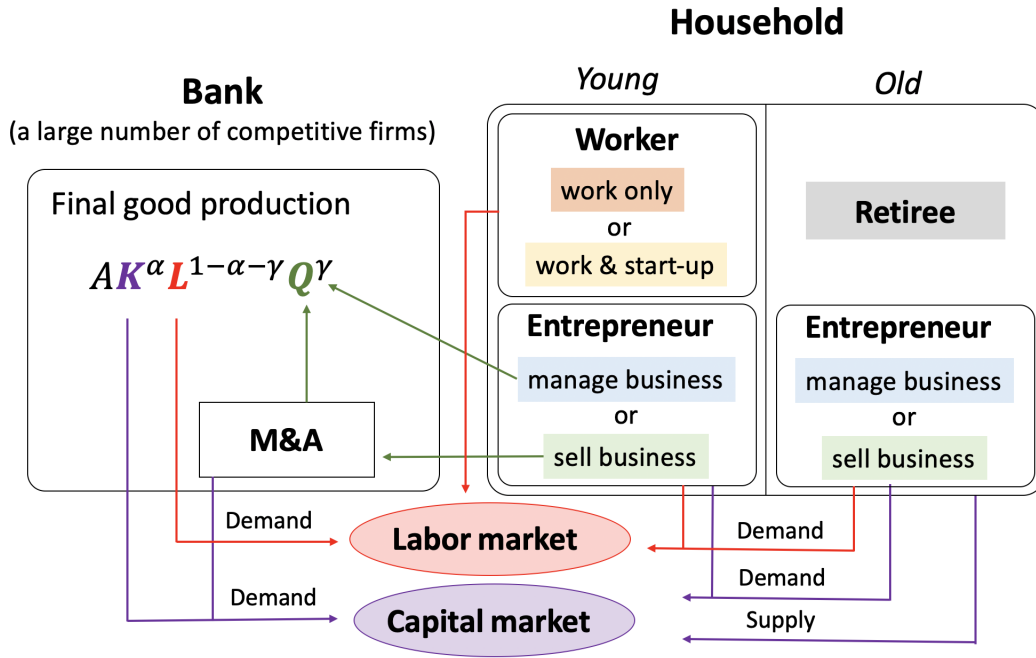


Figure 5: Quantitative model economy

3.1 Demographics and preferences

I adopt a life-cycle structure with intergenerational altruism, following [Cagetti and De Nardi \(2009\)](#), in which households pass through two stages of life: young and old. Each model period corresponds to one year. In each period, young households face a constant probability of aging, $(1 - \pi_y)$, while old households face a constant probability of dying, $(1 - \pi_o)$. Old households care about the utility of their offspring, who enter the economy upon the death of their parents and inherit the bequeathed assets and business. The probabilities of aging and death are calibrated so that the implied average lengths of working life and retirement are realistic. The population is normalized

to one. Households derive utility from consumption according to a CRRA utility function,

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}.$$

3.2 Heterogeneity

Households are heterogeneous in labor productivity $y \in \mathcal{Y}$, entrepreneurial idea $\theta \in \Theta$, private-business ownership and productivity $z \in \mathcal{Z}$, asset holdings $a \in \mathcal{A}$, and life stage $\iota \in \mathcal{I} = \{young, old\}$. The individual state is therefore

$$s = (z, a, y, \theta, \iota) \in \mathcal{S} \equiv \mathcal{Z} \times \mathcal{A} \times \mathcal{Y} \times \Theta \times \mathcal{I},$$

and let Ψ denote the cross-sectional distribution over $\mathcal{S}$.

Labor productivity. Labor productivity follows an AR(1) process in logs:

$$\log y_{t+1} = \rho_y \log y_t + \varepsilon_{y,t+1}, \quad \varepsilon_{y,t+1} \sim \mathcal{N}(0, \sigma_y^2), \quad (30)$$

with transition kernel $f(y' | y)$.

Entrepreneurial idea. Entrepreneurial idea θ follows an AR(1) process in logs:

$$\log \theta_{t+1} = \rho_\theta \log \theta_t + \varepsilon_{\theta,t+1}, \quad \varepsilon_{\theta,t+1} \sim \mathcal{N}(0, \sigma_\theta^2), \quad (31)$$

with transition kernel $g(\theta' | \theta)$.

Business ownership and productivity. Business ownership and productivity are summarized by the state variable z :

$$z = \begin{cases} z_0, & \text{if no business is owned,} \\ z_b \in \mathcal{Z}_b = \{z_1, z_2, \dots, z_N\}, & \text{if a business is owned, with productivity } z_b. \end{cases} \quad (32)$$

The state variable z evolves through both endogenous decisions, such as entry and sale, and exogenous shocks to business productivity. Figure 6 illustrates this evolution by showing the within-period timing of household decisions for young households.

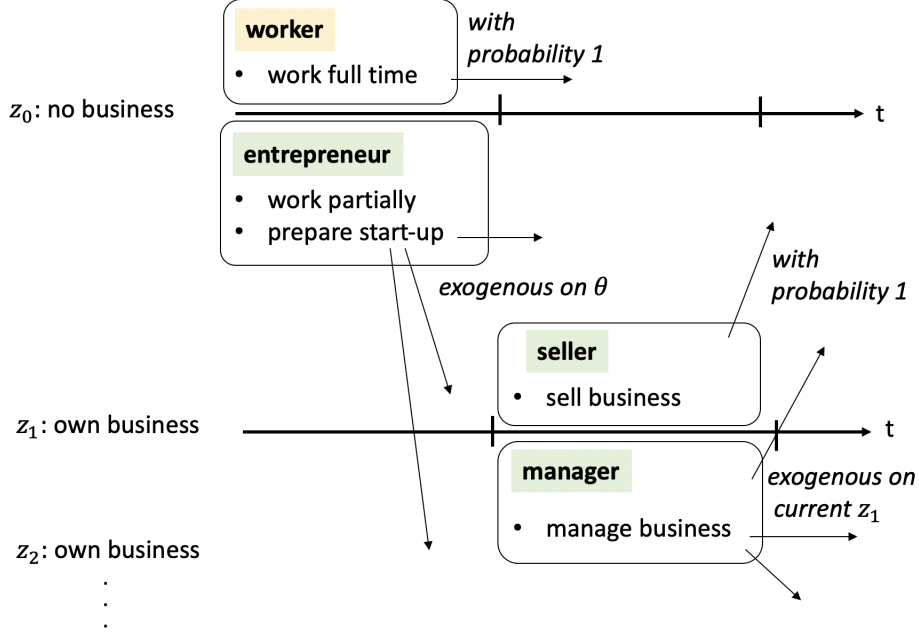


Figure 6: Timeline of household decisions within each period

Entry decision. In each period, households in the no-business state, $z = z_0$, choose whether to remain workers or to attempt to start a business. Households that remain workers stay in state z_0 in the next period. Prospective entrepreneurs pay a sunk entry cost and learn their initial business productivity at the beginning of the next period. Conditional on attempting entry, next-period productivity z' is drawn from the distribution $h_e(z' | \theta)$. I assume that the failure probability is constant across θ and that, conditional on successful entry, the probability of drawing business productivity is decreasing and convex in the productivity index. To capture this structure, I use an exponential specification and assume that a higher entrepreneurial idea θ shifts the startup-productivity distribution toward more productive outcomes:

$$h_e(z' | \theta) = \begin{cases} \delta_e, & \text{if } z' = z_0, \\ (1 - \delta_e) \frac{\exp(-\frac{1}{\theta}j)}{\sum_{m=1}^{N_z} \exp(-\frac{1}{\theta}m)}, & \text{if } z' = z_j, \quad j = 1, \dots, N_z. \end{cases} \quad (33)$$

Sell/continue decision. Households that already own a business, $z = z_b \in \mathcal{Z}_b$, choose either to sell the firm to the bank's M&A sector or to continue managing it. Those who sell transition to z_0 in the next period. If they continue operating the firm, they face a constant bankruptcy risk δ_c , and business productivity evolves according to an AR(1) process in logs:

$$\log z_{t+1} = \rho_z \log z_t + \varepsilon_{z,t+1}, \quad \varepsilon_{z,t+1} \sim \mathcal{N}(0, \sigma_z^2), \quad (34)$$

with transition kernel $h(z'_b \mid z_b)$. Hence, the transition kernel for z for households that own a business is

$$h_c(z' \mid z_b) = \begin{cases} \delta_c, & \text{if } z' = z_0, \\ (1 - \delta_c)h(z' \mid z_b), & \text{if } z' = z_j, \quad j = 1, \dots, N_z. \end{cases} \quad (35)$$

Assets. Asset holdings a are chosen endogenously in each period. I assume that $a' \in \mathcal{A}$, where $\mathcal{A}$ may incorporate a borrowing limit.

3.3 Technology

Intermediate-good production. Each entrepreneur who owns a business produces a differentiated intermediate good using capital k and labor ℓ ,

$$x_i = z_i k_i^\varphi \ell_i^{1-\varphi}. \quad (36)$$

When the bank's M&A sector acquires a business and operates it, output is given by

$$x_i = \lambda z_i k_i^\varphi \ell_i^{1-\varphi}, \quad 0 < \lambda \leq 1, \quad (37)$$

where λ captures the efficiency loss under bank control, or equivalently the nonpecuniary benefit that entrepreneurs derive from control, following [Chari et al. \(2005\)](#). Thus,

$$x = \begin{cases} z k^\varphi \ell^{1-\varphi}, & \text{if owner-managed,} \\ \lambda z k^\varphi \ell^{1-\varphi}, & \text{if bank-managed (M\&A).} \end{cases} \quad (38)$$

Final-good production. The bank's production sector produces the final good using capital K_c , labor L_c , and a CES aggregate of differentiated intermediate inputs, Q :

$$Y = AK_c^\alpha L_c^{1-\alpha-\gamma} Q^\gamma, \quad \alpha \in (0, 1), \gamma \in (0, 1 - \alpha). \quad (39)$$

Let Φ_t denote the distribution of firms operated by the bank's M&A division at the beginning of period t , before new business sales occur. When households sell their businesses to the bank, these newly acquired firms enter the post-trade bank-operated distribution in period t . Firms enter this distribution when households sell their businesses to the bank. For any measurable set $B \subseteq \mathcal{Z}_b$, define the inflow of newly acquired firms as

$$M_t(B) = \int_{\mathcal{S}} \mathbb{1}_s(s) \mathbf{1}\{z(s) \in B\} d\Psi_t(s), \quad (40)$$

where $\mathbb{1}_s(s)$ is an indicator equal to one if a household in state s sells its business in the current period.

The post-trade distribution of bank-managed firms is therefore

$$\tilde{\Phi}_t(B) = \Phi_t(B) + M_t(B). \quad (41)$$

After production in period t , all firms in the post-trade bank-operated distribution $\tilde{\Phi}_t$ evolve according to the transition kernel $h_c(z' | z)$, so the law of motion for Φ_t is

$$\Phi_{t+1}(B) = \int_{\mathcal{Z}_b} h_c(B | z_b) d\tilde{\Phi}_t(z_b), \quad B \subseteq \mathcal{Z}_b. \quad (42)$$

Because Φ_t is defined only on $\mathcal{Z}_b$, any mass that transitions to z_0 exits the bank-managed distribution automatically.

Let $\mathbb{1}_m(s)$ be an indicator equal to one if a household in state s operates its business in the current period, that is, chooses to manage rather than sell, and zero otherwise. The CES aggregator is

$$Q_t = \left(\int_{\mathcal{S}} \mathbb{1}_m(s) x_m(s)^\mu d\Psi_t(s) + \int_{\mathcal{Z}_b} x_b(z)^\mu d\tilde{\Phi}_t(z) \right)^{1/\mu}, \quad \mu \in (0, 1]. \quad (43)$$

Final-good firm's problem and factor prices. A representative competitive final-good firm chooses inputs taking prices as given. Let w denote the wage per efficiency unit and r the risk-free interest rate. The user cost of capital is $r + \delta$, where δ is the depreciation rate. Profit maximization implies

$$w = (1 - \alpha - \gamma) AK_c^\alpha L_c^{-\alpha-\gamma} Q^\gamma, \quad (44)$$

and

$$r + \delta = \alpha AK_c^{\alpha-1} L_c^{1-\alpha-\gamma} Q^\gamma. \quad (45)$$

Moreover, the implied inverse demand schedule for any differentiated intermediate good of quantity x is

$$\xi(x) = \gamma AK_c^\alpha L_c^{1-\alpha-\gamma} Q^{\gamma-\mu} x^{\mu-1}, \quad (46)$$

where $\xi(x)$ is the price per unit of the intermediate good. Individual intermediate producers take (K_c, L_c, Q) as given and therefore face a downward-sloping demand schedule when $\mu < 1$.

3.4 Households

Households' decision problems depend on both their life stage and whether they currently own a business. Accordingly, I define separate value functions by life stage and business status: (i) households without a business, $(z = z_0)$, and (ii) households with a business, $(z = z_b \in \mathcal{Z}_b)$. During the young stage, households may attempt to start a business. Once they become old, however, they retire unless they are business owners, following [Cagetti and De Nardi \(2006\)](#). This assumption is intended to capture the fact that, whereas most workers retire before age 65, entrepreneurs often continue operating their businesses until much later.

3.4.1 Young households problem

Young households without a business. A household without a business chooses between working and attempting entry:

$$V(z_0, a, y, \theta) = \max \{V_w(z_0, a, y, \theta), V_e(z_0, a, y, \theta)\}. \quad (47)$$

Worker. Workers earn labor income wy and asset income ra . Let τ_w denote the labor-income

tax rate, τ_k the capital-income tax rate, and τ_c the consumption tax rate. Regardless of whether they age in the next period, they remain in the no-business state z_0 . Conditional on remaining young, the expected continuation value is taken over (y', θ') conditional on (y, θ) using $f(y' | y)$ and $g(\theta' | \theta)$. If the worker becomes old, the household retires, and the continuation value is $W_r(z_0, a')$. The worker's problem is

$$V_w(z_0, a, y, \theta) = \max_{c, a' \geq 0} \{u(c) + \beta \pi_y \mathbb{E}[V(z_0, a', y', \theta')] + \beta(1 - \pi_y) W_r(z_0, a')\}, \quad (48)$$

subject to

$$a' = (1 + r(1 - \tau_k))a + (1 - \tau_w)wy - (1 + \tau_c)c. \quad (49)$$

Startup entrepreneur. If the household attempts to start a business, it pays a sunk entry cost κ and supplies full-time labor to wage work during the startup phase. This assumption is motivated by evidence that about half of new entrepreneurs maintain full-time employment elsewhere while starting their businesses, and that the majority of entrepreneurs continue to engage in some paid work (Campbell and De Nardi, 2009). Next-period business productivity z' is drawn from $h_e(z' | \theta)$. Hence, conditional on remaining young, the expected continuation value is taken over (z', y', θ') conditional on (y, θ) using $h_e(z' | \theta)$, $f(y' | y)$, and $g(\theta' | \theta)$. If the household becomes old, the continuation value is $W(z', a')$, where the expectation is taken over z' conditional on θ using $h_e(z' | \theta)$. The startup entrepreneur's problem is

$$V_e(z_0, a, y, \theta) = \max_{c, a' \geq 0} \{u(c) + \beta \pi_y \mathbb{E}[V(z', a', y', \theta')] + \beta(1 - \pi_y) \mathbb{E}[W(z', a')]\}, \quad (50)$$

subject to

$$a' = (1 + r(1 - \tau_k))a + (1 - \tau_w)wy - (1 + \tau_c)c - \kappa. \quad (51)$$

Young households with a business. A young business owner chooses whether to continue managing the business or to sell it:

$$V(z_b, a, y, \theta) = \max \{V_m(z_b, a, y, \theta), V_s(z_b, a, y, \theta)\}. \quad (52)$$

Manager. A managing entrepreneur chooses capital k , labor ℓ , consumption, and savings.

Period profits are given by

$$\pi(z, k, \ell) = \xi(x) x - (r + \delta)k - w\ell, \quad x = zk^\varphi \ell^{1-\varphi}. \quad (53)$$

Capital investment is subject to a collateral constraint that depends on the entrepreneur's assets and business productivity:

$$k \leq \psi(z) a, \quad (54)$$

where $\psi(\cdot)$ governs the tightness of borrowing constraints. Conditional on remaining young, the expected continuation value is taken over (z', y', θ') conditional on (z_b, y, θ) using $h_c(z' | z_b)$, $f(y' | y)$, and $g(\theta' | \theta)$. If the entrepreneur becomes old, the expected continuation value is taken over z' conditional on z_b using $h_c(z' | z_b)$. The value function for a managing entrepreneur is

$$V_m(z_b, a, y, \theta) = \max_{c, k, \ell, a' \geq 0} \left\{ u(c) + \beta \pi_y \mathbb{E}[V(z', a', y', \theta')] + \beta(1 - \pi_y) \mathbb{E}[W(z', a')] \right\}, \quad (55)$$

subject to (54) and

$$a' = (1 + r(1 - \tau_k)) a + (1 - \tau_k) \pi(z_b, k, \ell) - (1 + \tau_c) c. \quad (56)$$

Seller. If the entrepreneur sells the business, she pays a fixed cost ϕ , and the business is transferred to the bank's M&A sector at price $p(z_b)$. Under realization-based taxation, the seller pays a capital-gains tax at rate τ_g upon sale. Because the entrepreneur sells the business in the current period, she becomes a non-business owner in the next period. Conditional on remaining young, the expected continuation value is taken over (y', θ') conditional on (y, θ) using $f(y' | y)$ and $g(\theta' | \theta)$. If she becomes old, she retires. The seller's problem is

$$V_s(z_b, a, y, \theta) = \max_{c, a' \geq 0} \left\{ u(c) + \beta \pi_y \mathbb{E}[V(z_0, a', y', \theta')] + \beta(1 - \pi_y) W_r(z_0, a') \right\}, \quad (57)$$

subject to

$$a' = (1 + r(1 - \tau_k)) a + (1 - \tau_g) p(z_b) - \phi - (1 + \tau_c) c. \quad (58)$$

3.4.2 Old households problem

Old households that own a business can choose either to continue managing it or to sell it and retire, whereas old households without a business are retirees.

Old households without a business (retirees). Retirees receive pension income T_p in every period and remain retirees for as long as they survive. Upon death, their offspring enter the economy with the bequeathed assets, and their labor productivity y and entrepreneurial idea θ are drawn from the invariant distribution. Old households value the utility of their offspring with weight η . The retiree's value function is

$$W_r(z_0, a) = \max_{c, a' \geq 0} \{u(c) + \beta \pi_o W_r(z_0, a') + \eta \beta (1 - \pi_o) \mathbb{E}V(z_0, a', y', \theta')\} \quad (59)$$

subject to

$$a' = T_p + (1 + r(1 - \tau_k))a - (1 + \tau_c)c. \quad (60)$$

Old households with a business. An old business owner chooses whether to continue managing the business or to sell it:

$$W(z_b, a) = \max\{W_m(z_b, a), W_s(z_b, a)\}. \quad (61)$$

Manager. An old entrepreneur who continues to manage the business draws next-period business productivity conditional on z_b according to $h_c(z' | z_b)$. Upon death, the offspring enter the economy with bequeathed assets a' and inherit the business, whose productivity z' is drawn conditional on the parent's business productivity z_b using $h_c(z' | z_b)$. The offspring's labor productivity y' and entrepreneurial idea θ' are drawn from the invariant distribution. The value function is

$$W_m(z_b, a) = \max_{c, k, \ell, a' \geq 0} \{u(c) + \beta \pi_o \mathbb{E}W(z', a') + \eta \beta (1 - \pi_o) \mathbb{E}V(z', a', y', \theta')\} \quad (62)$$

subject to (54) and (56).

Seller. An old entrepreneur who sells the business in the current period becomes a retiree in the next period. Upon death, the offspring enter the economy with the bequeathed assets, and their labor productivity y' and entrepreneurial idea θ' are drawn from the invariant distribution.

The seller's problem is

$$W_s(z_b, a) = \max_{c, a' \geq 0} \{u(c) + \beta \pi_o W_r(z_0, a') + \eta \beta (1 - \pi_o) \mathbb{E}V(z_0, a', y', \theta')\} \quad (63)$$

subject to (58).

3.5 Bank

The bank has two divisions: final-good production and M&A. Final-good production was described above. The M&A division acquires businesses and operates them with efficiency wedge λ .

M&A division and the purchase-price function. Let $\pi_b(z, k, \ell)$ denote the period profit earned when the bank operates an acquired firm with productivity z using capital k and labor ℓ :

$$\pi_{\text{bank}}(z, k, \ell) = \xi(x) x - (r + \delta)k - w\ell, \quad x = \lambda z k^\varphi \ell^{1-\varphi}. \quad (64)$$

The M&A division purchases a business at price $p(z)$, defined as the present value of after-tax profits from operating the acquired firm. The price function satisfies

$$p(z_b) = \max_{k, \ell \geq 0} \left\{ (1 - \tau_k) \pi_{\text{bank}}(z_b, k, \ell) + \frac{1}{1 + r} \int_{\mathcal{Z}} p(z') dh_c(z' | z_b) \right\}, \quad (65)$$

with boundary condition

$$p(z_0) = 0. \quad (66)$$

3.6 Government

The government collects taxes on consumption, labor income, capital income, and realized capital gains. Let $\mathbb{1}_w(s)$ and $\mathbb{1}_e(s)$ indicate whether a household without a business works or attempts entry in the current period, respectively. Let $\mathbb{1}_m(s)$ and $\mathbb{1}_s(s)$ indicate whether a business owner manages or sells in the current period, respectively. Let $\mathbb{1}_r(s)$ indicate whether a household is retired. In the baseline analysis, all tax revenue is used to finance government purchases G .

The government budget constraint is

$$G = \int_{\mathcal{S}} \left[\tau_c c(s) + \tau_k r a + \mathbb{1}_w(s) \tau_w w y + \mathbb{1}_e(s) \tau_w w y + \mathbb{1}_s(s) \tau_g p(z_b) + \mathbb{1}_m(s) \tau_k \pi(z_b, k(s), \ell(s)) \right] d\Psi(s) \\ + \int_{\mathcal{Z}_b} \tau_k \pi_{\text{bank}}(z_b, k(z_b), \ell(z_b)) d\Phi(z) - \int_{\mathcal{S}} \mathbb{1}_r(s) T_p. \quad (67)$$

3.7 Market clearing

Capital demanded by the final-good producer and by intermediate producers must equal aggregate asset holdings. The capital-market clearing condition is

$$K_c + \int_{\mathcal{S}} \mathbb{1}_m(s) k_m(s) d\Psi(s) + \int_{\mathcal{Z}_b} k_b(z) d\tilde{\Phi}(z) = \int_{\mathcal{S}} a d\Psi(s). \quad (68)$$

Labor demand by the final-good producer and by intermediate producers must equal the total efficiency units supplied by workers and startup entrepreneurs:

$$L_c + \int_{\mathcal{S}} \mathbb{1}_m(s) \ell(s) d\Psi(s) + \int_{\mathcal{Z}_b} \ell(z) d\tilde{\Phi}(z) = \int_{\mathcal{S}} [\mathbb{1}_w(s) + \mathbb{1}_e(s)] y d\Psi(s). \quad (69)$$

3.8 Equilibrium

A stationary equilibrium consists of prices (r, w) ; aggregate inputs (K_c, L_c, Q) ; household policy functions for consumption and saving $\{c(s), a'(s)\}_{s \in \mathcal{S}}$; discrete occupational and sale choices encoded by $\{\mathbb{1}_w(s), \mathbb{1}_e(s), \mathbb{1}_m(s), \mathbb{1}_s(s), \mathbb{1}_r(s)\}_{s \in \mathcal{S}}$; entrepreneurial capital and labor choices $\{k(s), \ell(s)\}_{s \in \mathcal{S}}$; bank policies $\{k(z_b), \ell(z_b)\}_{z_b \in \mathcal{Z}_b}$; a purchase-price function $p(z_b)$; and invariant distributions Ψ^* on $\mathcal{S}$ and Φ^* on $\mathcal{Z}_b$ such that:

1. Given prices and the demand schedule (46), households solve the dynamic programming problems in (47)–(63).
2. The final-good firm optimally chooses inputs, and factor prices satisfy (44)–(45); intermediate-good prices satisfy (46).
3. The M&A purchase-price function satisfies (65)–(66).
4. The government budget constraint (67) holds.

5. Markets clear, so (68) and (69) hold.
6. The distributions Ψ^* and Φ^* are invariant under the induced policy rules and exogenous transition kernels.

4 Calibration

Table 1 reports the parameter values used in the quantitative model. Some parameters are calibrated externally taken from the literature or direct empirical counterparts, while the remaining parameters are calibrated internally to match targeted moments.

Parameter	Value	Source / Target
<i>Externally calibrated</i>		
α	0.07	NIPA data, average 2010–2019
γ	0.35	Giandrea and Sprague (2017)
φ	0.605	SOI data, average 2015–2019
μ	0.77	standard value in literature
δ	0.05	standard value in literature
τ_k, τ_g	0.234	McDaniel (2007), average 2010–2019
τ_w	0.204	McDaniel (2007), average 2010–2019
τ_c	0.075	McDaniel (2007), average 2010–2019
T_p	40% of average yearly income	Cagetti and De Nardi (2006)
ρ_y	0.95	Storesletten et al. (2004)
σ_y	0.255	Heathcote et al. (2023)
π_y	0.978	45 years of average young period
π_o	0.929	14 years of average old period
κ	25% of average yearly income	Campbell and De Nardi (2007)
δ_e	0.18	Fairlie (2022)
δ_c	0.08	BDS data, average 2010–2019
η	1.0	Cagetti and De Nardi (2006)
σ	1.5	Attanasio et al. (1999)
<i>Internally calibrated</i>		
β	0.956	Capital-to-GDP ratio
λ	0.76	Business-value held by entrepreneurs to total wealth
$\bar{\psi}$	0.40	Private-business debt-to-GDP ratio
ρ_z	0.45	Entrepreneur share at top 1% non-business wealth
σ_z	0.34	Gini coefficient of wealth other than business value
ρ_θ	0.90	Entry rate of startup
σ_θ	0.45	Entrepreneur population share
ϕ	4.15	Seller share among entrepreneurs

Table 1: Externally and internally calibrated parameters (annual frequency).

4.1 External calibration

Production related parameters. In the final goods production function, the capital share α is set at 7 percent—the observed ratio of corporate profits (non-financial domestic corporation) to GDP. The intermediate-goods share γ is calibrated so that the labor share $1 - \alpha - \gamma$ equals 58

percent, consistent with the estimate of [Giandrea and Sprague \(2017\)](#). The capital share φ in the private business production is set at 0.605, so that labor share $1 - \varphi$ is equal to the average labor cost ratio to value added in S corporation statistics recorded in the Statistics of Income (SOI) Tax Stats during 2015-2019. The curvature parameter of the CES aggregator for intermediate goods production, μ , is set to 0.77. This value lies between $\mu = 0.75$ and $\mu = 0.8$, which imply markups of 1.33 and 1.25, respectively. Because aggregate markup estimates vary widely in the literature, I choose an intermediate value within the standard range used in the firm-dynamics literature. As is standard in literature, capital depreciation rate δ is set to 0.05.

Tax rates and pension transfer. Each tax rates τ_k , τ_g , τ_w , and τ_c are set at 25 percent, 25 percent, 22.4 percent, and 7.5 percent respectively, which is estimated by [McDaniel \(2007\)](#). The pension transfer is 40% of average labor income following [Cagetti and De Nardi \(2006\)](#).

Labor productivity. Labor productivity follows AR(1) process. I set $\rho_y = 0.95$ following [Storesletten et al. \(2004\)](#). As the households in the model choose the occupation, I internally calibrate the σ_y to match the average Gini coefficient of hourly wage during 2010-2019 estimated by [Heathcote et al. \(2023\)](#).

Probability of aging and of death. The probability of aging and of death are set following [Cagetti and De Nardi \(2006\)](#) so that the average length of young period is 45 years (age 20-65) and the average length of old period is 14 years (age 65-79) where age of 79 is the average life expectancy in U.S.

Private firm dynamics. The entry sunk cost for start up entrepreneurs are set to 25% of average labor income as calculated by the average money investment for entry by new entrepreneurs reported in [Campbell and De Nardi \(2007\)](#) and average of wage and salary income using the Current Population Survey (CPS) data. The death shock by newly entered private business is set to 18% ([Fairlie \(2022\)](#)) and the death shock by incumbent private business is set to 8% from the average firm death rate from Business Dynamics Statistics data during period of 2010-2019 .

Intergenerational altruism and preference. I assume perfect intergenerational altruism as in [Cagetti and De Nardi \(2006\)](#). The curvature of CRRA utility function is set to 1.5 following [Attanasio et al. \(1999\)](#).

4.2 Internal calibration

The following parameters are calibrated jointly within the model:

- the discount factor β ;
- the bank's production-inefficiency parameter λ ;
- the collateral schedule $\psi(z)$ in the constraint $k \leq \psi(z)a$. Following [Rotberg and Steinberg \(2024\)](#), in the discretized model I impose

$$\psi(z_n) = 1 + \bar{\psi} \frac{n-1}{N-1}, \quad z_n \in \{z_1, \dots, z_N\};$$

- the persistence coefficient and standard deviation (ρ_z, σ_z) in the process for z ;
- the persistence coefficient and standard deviation $(\rho_\theta, \sigma_\theta)$ in the process for θ ;
- the fixed cost for selling private business ϕ .

These parameters are chosen so that the model matches the targeted moments in [Table 2](#): (i) the capital-to-GDP ratio; (ii) the business value held by entrepreneurs to total wealth in economy; (iii) the private-business debt-to-GDP ratio; (iv) the gini coefficient of non-business wealth; (v) the population share by self-employed business owners who actively manage their business at top 1% non-business wealth position; (vi) the entry rate of starting up business; (vii) the entrepreneur population share; (viii) the sellers share among entrepreneurs.

4.3 Targeted moments

[Table 2](#) compares the empirical moments used for internal calibration with their model counterparts. Overall, the model matches the targeted moments reasonably well.

Target	Data	Model	Source
Aggregate capital to GDP	3.00	3.04	BEA, average 2010–2019
Business value to total wealth	0.19	0.18	SCF, average 2010–2019
Private-business debt-to-GDP	0.11	0.13	SOI, average 2015–2019
Entrepreneur share at top 1%	0.35	0.30	SCF, average 2010–2019
Gini coefficient of non-business asset	0.84	0.84	SCF, average 2010–2019
New-entrepreneur rate	0.02	0.01	Cagetti and De Nardi (2009)
Entrepreneur population share	0.07	0.07	SCF, average 2010–2019
Sellers share	0.03	0.03	Bhandari et al. (2024)

Notes: Private business debt is calculated using total liability by non-financial business taking pro rate for S-corporations.

Table 2: Targeted moments for internal calibration

4.4 Non targeted moments

Table 3 reports non-targeted moments. Although the model does not match every moment perfectly, it captures several important patterns reasonably well, including the relative size of the wage bill of acquiring firms in the M&A sector compared with that of sellers before sale, bequests, the wealth distribution, occupational patterns, and tax revenues.

Target	Data	Model	Source
Relative size of wage bill			
25th percentile	0.90	0.65	Bhandari et al. (2024)
50th percentile	2.33	5.16	Bhandari et al. (2024)
75th percentile	13.5	7.93	Bhandari et al. (2024)
Bequests to total non-business wealth	0.01	0.01	Nishiyama (2000)
Gini coefficient of networth ($a + p(z)$)	0.89	0.85	SCF, average 2010–2019
Non-business asset share(a)			
by top 10%	0.73	0.68	SCF, average 2010–2019
by bottom 50%	0.01	0.00	SCF, average 2010–2019
Entrepreneur’s non-business wealth share	0.24	0.34	SCF, average 2010–2019
Average wealth held by entrepreneur to workers	4.65	5.80	SCF, average 2010–2019
Tax revenue to GDP	0.26	0.24	OECD, average 2015–2019
Consumption	0.04	0.05	OECD, average 2015–2019
Labor income	0.13	0.14	OECD, average 2015–2019
Capital income	0.04	0.04	OECD, average 2015–2019
Capital gains	0.009	0.007	OECD, average 2015–2019

Notes: Capital income tax revenue is calculated as the sum of (1) taxes on individual income and profits, multiplied by a pro-rated capital share of 0.3, and (2) taxes on corporate income, profits, and capital gains. When property taxes are also included, the capital-income-tax-revenue-to-GDP ratio increases to 7.7%.

Table 3: Non targeted moments for internal calibration

5 Model with an accrual-based capital-gains tax

This section modifies the model to incorporate an accrual-based (mark-to-market) capital-gains tax. Under accrual taxation, an entrepreneur pays taxes on changes in the market value of the

business while it is held. If the business value falls, the entrepreneur receives a refund (or an offset). For tractability, I assume full loss offset with contemporaneous rebates.

5.1 State augmentation and accrual tax liability

To implement accrual taxation, I augment the household state with a lagged business-productivity index $z^- \in \mathcal{Z}$, which records the previous period's business productivity if the household held a business then, and equals z_0 otherwise. The household state becomes

$$\tilde{s} = (z, a, y, \theta, z^-) \in \tilde{\mathcal{S}} \equiv \mathcal{Z} \times \mathcal{A} \times \mathcal{Y} \times \Theta \times \mathcal{Z}.$$

Let $p(z)$ denote the assessed market value of a business with productivity z , with normalization $p(z_0) = 0$. The accrual-based capital-gains tax liability is

$$T_g(z, z^-) \equiv \tau_g(p(z) - p(z^-)), \quad (70)$$

which can be negative (a rebate) when $p(z) < p(z^-)$. In budget constraints below, this enters as a payment $-T_g(z, z^-)$.

The transition matrix of z^- is

$$F(z^{-'} | z^-) = \begin{cases} z^{-'} = z \text{ for all occupation except seller} \\ z^{-'} = z_0 \text{ for seller} \end{cases} \quad (71)$$

5.2 Discussion: practical implementation

Policy proposals for accrual-based taxation (mark-to-market taxation) recognize that annual valuation of privately held businesses is difficult. In practice, such reforms would likely allow infrequent appraisals and impute interim appreciation using an interest-rate benchmark, and they may permit installment payment of tax liabilities to mitigate liquidity constraints. Loss offsets are also typically implemented via carry-forward rules. For tractability, I abstract from these institutional details and model annual accrual taxation with full contemporaneous loss refunds as in (70).

6 Results

This section conducts counterfactual experiments comparing three alternative capital-gains tax regimes: (i) a *realization-based* capital-gains tax; (ii) an *accrual-based* capital-gains tax; and (iii) a *hybrid* regime that combines the two. In the hybrid regime, accrual-based taxation applies only to wealthy households, while the rest of the population faces realization-based taxation. This hybrid design follows recent policy proposals, including the proposal advanced by the Biden administration.

6.1 Realization-based taxation vs. Accrual-based taxation

I first examine the two main mechanisms emphasized by the stylized model: firm-entry deterrence and capital lock-in. I then discuss how these mechanisms affect aggregate quantities, inequality, and welfare.

Firm entry. The stylized model implies that accrual-based taxation discourages firm entry. The mechanism is that, upon successful business creation, a start-up entrepreneur must pay the accrual tax, which reduces the net payoff from entering entrepreneurship. As a result, only households with sufficiently high business-idea ability choose to become entrepreneurs, and the number of entrants falls.

This mechanism is also present in the quantitative model. Table 4 shows that switching to accrual-based taxation reduces the population of start-up entrepreneurs by 3.4 percent, while increasing their average business-idea ability by 8.1 percent.

Capital lock-in. The stylized model also shows that a realization-based capital-gains tax induces a lock-in effect: entrepreneurs who would otherwise sell their businesses choose instead to retain and manage them in order to defer capital-gains taxation. Switching to accrual-based taxation reduces this lock-in distortion because the tax liability no longer depends on whether the business is transferred. As a result, accrual-based taxation induces high-productivity business owners to sell their businesses. The number of sellers rises, while the number of managers falls.

Because high-productivity business owners are more likely to sell, the average productivity of the remaining managers declines. Table 4 shows that this mechanism from the stylized model also holds in the quantitative model. The number of managers falls by 7.4 percent, while the number of

sellers increases by 8.1 percent. The average business productivity of managers falls by 0.9 percent relative to the realization-based regime. In addition, because accrual-based taxation encourages high-productivity managers to sell their businesses and enhances business turnover, the average productivity of firms operated by the M&A sector increases by 0.2 percent.

Table 4: Firm Entry and Capital Lock-in under Accrual-based Taxation

Policy	Firm entry		Capital lock-in			
	Δ No. of entrants	Δ avg. θ of entrants	Δ No. of managers	Δ No. of sellers	Δ avg. z of managers	Δ avg. z of M&A firms
Accrual-based	-3.4	+8.1	-7.4	+8.1	-0.9	+0.2

Notes: All entries are percentage changes relative to the realization-based regime.

Aggregate outcomes. The two mechanisms described above have offsetting implications for aggregate production. Accrual-based taxation mitigates the lock-in distortion and encourages business turnover, but it also discourages firm entry by reducing the after-tax payoff from successful entrepreneurship. Table 5 summarizes the macroeconomic outcomes under accrual-based taxation relative to the realization-based regime.

Entrepreneurial production falls by 1.5 percent under accrual-based taxation. This decline reflects the fact that the entry-deterrence effect dominates the reduction in lock-in. In particular, production by M&A firms increases by 9.3 percent, while production by managers falls by 12.8 percent. As entrepreneurial production declines, capital demand falls, and the aggregate capital stock decreases by 0.7 percent. At the same time, households that are discouraged from entering entrepreneurship instead supply labor, increasing aggregate labor supply by 0.4 percent. Consequently, aggregate final-good production falls by 0.4 percent.

Table 5: Aggregate Quantity Outcomes under Accrual-based Taxation

Policy	Y	Q	x by managers	x by M&A firms	K	L	C
Accrual-based	-0.4	-1.5	-12.8	+9.3	-0.7	+0.4	-0.6

Notes: All entries are percentage changes relative to the realization-based regime.

Inequality and welfare. Table 6 reports the implications for inequality and welfare. The Gini coefficient of non-business assets decreases because entry into high-earning entrepreneurial occupations becomes more difficult. In contrast, the Gini coefficient of net worth increases because

only more productive entrepreneurs remain in business, and these entrepreneurs hold businesses with higher values. Overall, consumption-equivalent welfare falls by 0.4 percent when the economy switches from realization-based to accrual-based taxation.

Table 6: Inequality and Welfare

Policy	Gini coefficient		Consumption-equivalent welfare
	Non-business assets	Net worth	
Realization-based	0.843	0.886	–
Accrual-based	0.839	0.940	–0.4%

Notes: Consumption-equivalent welfare is measured relative to the realization-based regime.

Table 7 reports consumption-equivalent welfare changes relative to the realization-based status quo, separately by exogenous household characteristics: age, working ability, and business idea ability. Switching to accrual-based taxation reduces welfare more for old households than for young households. This is because old households are more likely to be incumbent entrepreneurs, who are directly affected by accrual-based taxation. In the model, collateral constraints require households to accumulate sufficient assets before operating a business, making entrepreneurship more common among older households.

In terms of working ability, welfare losses are larger for low- and high-ability households than for medium-ability households. Low-ability households have a lower comparative advantage in working and are therefore more likely to become entrepreneurs. As a result, they are more exposed to accrual-based taxation and experience larger welfare losses. High-ability households, by contrast, have a stronger comparative advantage in working. Since the firm-entry deterrence effect increases labor supply and lowers wages, these households are also negatively affected through the decline in labor income.

A similar pattern appears across business idea ability. Households with low and high business idea ability experience larger welfare losses than those with medium ability. Households with low business idea ability are hurt because they are more likely to work, and the policy-induced increase in labor supply lowers wages. Households with high business idea ability are more likely to be incumbent entrepreneurs and therefore bear a larger tax burden when business values are taxed on an accrual basis.

Table 7: Consumption-equivalent welfare by exogenous state

Policy	Age		Working ability y			Business idea ability θ		
	Young	Old	Low	Medium	High	Low	Medium	High
Accrual-based	-0.36	-0.42	-0.68	-0.32	-0.40	-0.43	-0.31	-0.76

Notes: Consumption-equivalent welfare is measured relative to the realization-based regime.

6.2 Hybrid regime of accrual-based taxation for the wealthy

This subsection examines a hybrid tax regime in which accrual-based taxation applies only to wealthy households, while the rest of the population remains subject to realization-based taxation. I consider different wealth cutoff levels above which households are subject to accrual-based taxation.

The occupation distribution suggests that wealthy households are more likely to be incumbent entrepreneurs. Therefore, applying accrual-based taxation only to wealthy households may generate a smaller firm-entry deterrence effect. However, if accrual-based taxation applies only to a very small share of the population, it may also have a limited effect on reducing capital lock-in. Hence, the choice of the cutoff level determines which of the two mechanisms dominates. I first present the results for these two mechanisms and then discuss the aggregate outcomes.

Intermediate goods production. Figure 7 shows the productivity of businesses operated by the M&A sector and by managers, as well as the populations of sellers and managers, under different wealth cutoffs for accrual-based taxation. All variables are reported relative to the realization-based status quo.

When accrual-based taxation applies only to very wealthy households, the firm-entry deterrence effect is mitigated. As a result, the population of managers and their average productivity decline less. On the other hand, applying accrual-based taxation to only a small share of the population has a limited effect on reducing capital lock-in and encouraging business turnover. Thus, the population of sellers and the productivity of businesses operated by the M&A sector increase only modestly. The overall effect on intermediate goods production therefore depends crucially on the choice of the cutoff level.

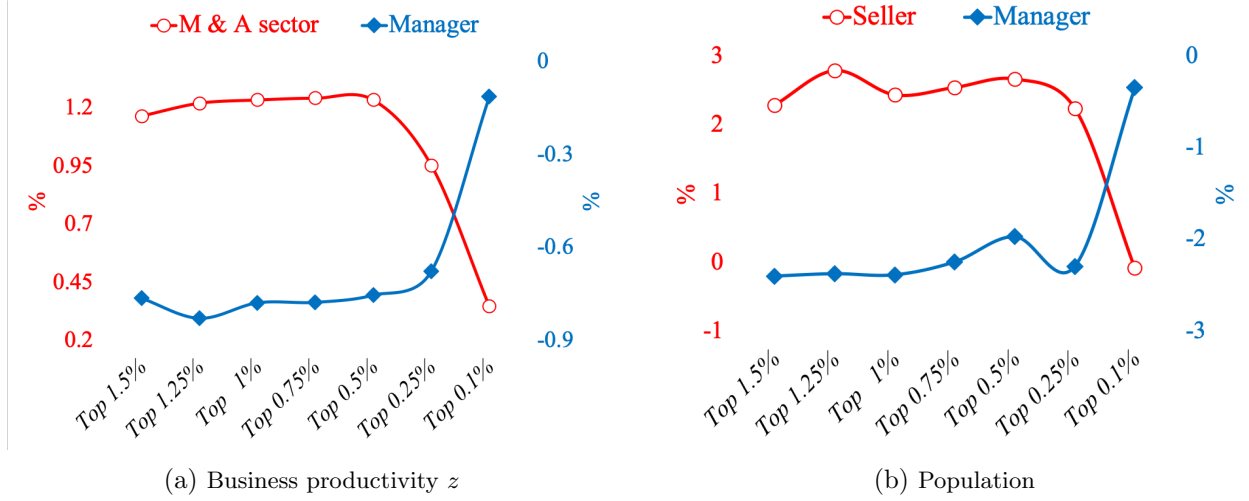


Figure 7: Firm productivity and population under hybrid accrual-based taxation

Figure 8 shows aggregate intermediate goods production and sectoral production by the M&A sector and managers. As the population subject to accrual-based taxation expands, the firm-entry deterrence effect becomes stronger and reduces production by managers. At the same time, the reduction in capital lock-in becomes stronger, increasing production by the M&A sector. As a result, aggregate intermediate goods production exhibits a hump-shaped pattern, with the top 0.5% cutoff generating the largest increase in production.

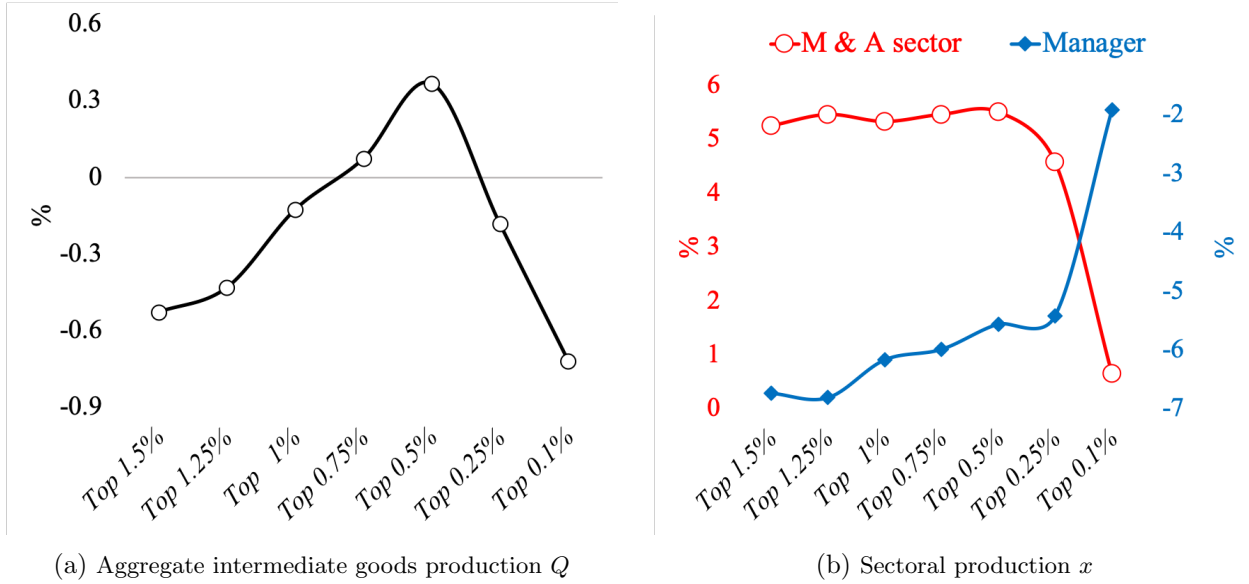


Figure 8: Intermediate goods production under hybrid accrual-based taxation

Final good production and aggregate welfare. Figure 9 shows aggregate final-good production and consumption-equivalent welfare relative to the realization-based status quo. Both variables exhibit a hump-shaped pattern, similar to aggregate intermediate goods production. This result reflects the same trade-off: a broader application of accrual-based taxation reduces capital lock-in more strongly, but it also increases the firm-entry deterrence effect.

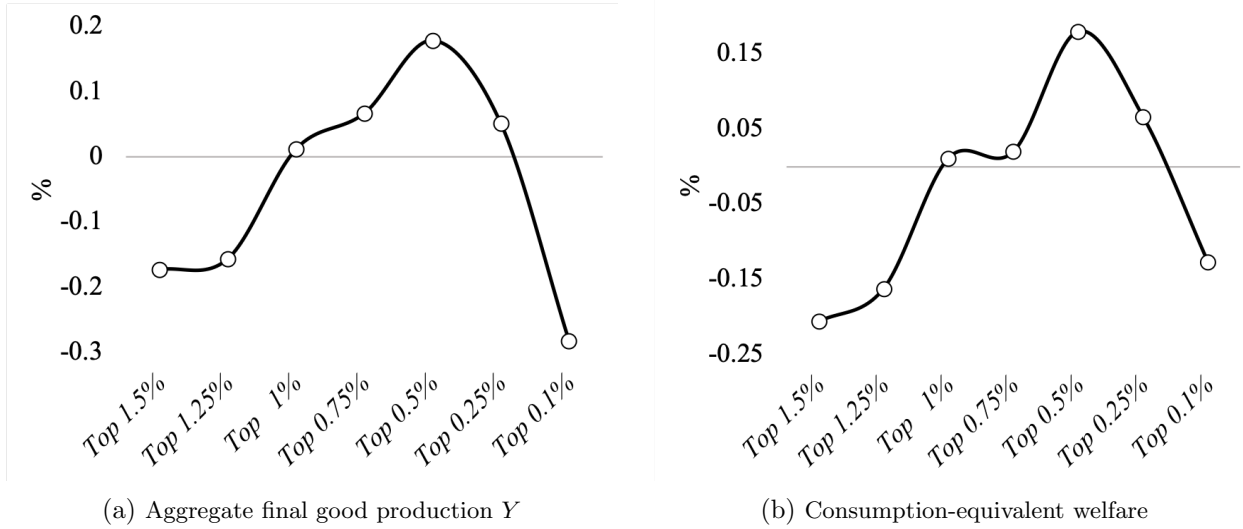


Figure 9: Final good production and welfare under hybrid accrual-based taxation

Consumption-equivalent welfare by exogenous state. Figure 10 illustrates consumption-equivalent welfare relative to the realization-based status quo by exogenous household characteristics. The comparison by age shows a pattern similar to the full accrual-based regime: old households are affected more negatively than young households.

However, under the hybrid regime, high-working-ability households experience welfare gains, whereas they were negatively affected under economy-wide accrual-based taxation. This is because taxing very wealthy entrepreneurs reduces their capital holdings and raises the return on capital. High-working-ability households benefit from the higher return on savings. In addition, for cutoff levels between the top 1.5% and the top 0.5%, households with high business idea ability experience welfare gains. This pattern is consistent with the response of intermediate goods production.

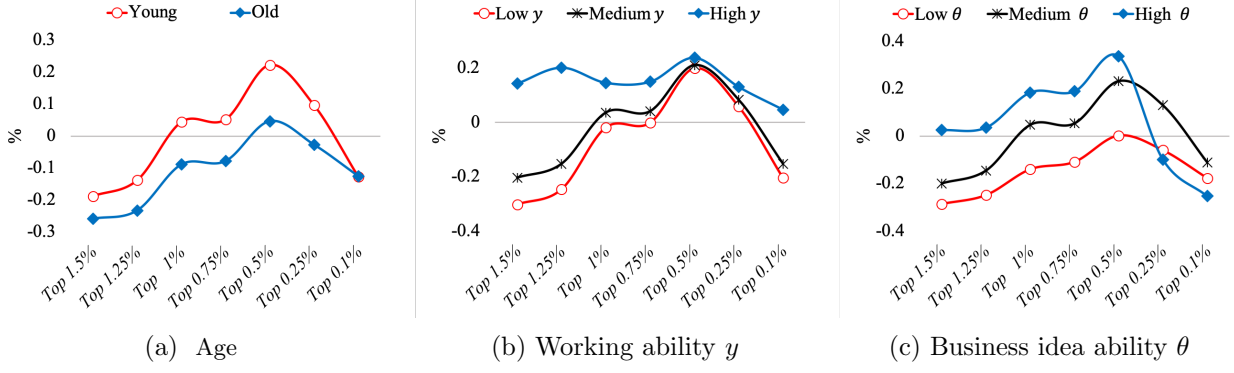


Figure 10: Consumption-equivalent welfare by exogenous state

7 Conclusion

This paper studies the macroeconomic and distributional consequences of replacing realization-based capital-gains taxation with an accrual-based alternative, focusing on capital gains from privately held businesses. The quantitative analysis shows that an economy-wide shift to accrual-based taxation can reduce aggregate production because entry deterrence dominates the reduction in capital lock-in. However, a targeted reform that applies accrual taxation only to very wealthy households can dampen the entry-deterrence margin while still mitigating lock-in, generating gains in output and an increase in aggregate welfare.

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8 Appendix

A. Proof of Proposition 1.

Proof. Fix (w, r) and let $\beta \equiv (1+r)^{-1}$. In the baseline (no capital-gains tax) economy, the relevant value functions at $z = 0$ admit closed-form expressions.

Step 1: Closed-form value functions. The worker value is

$$V^w = \frac{w}{1-\beta}.$$

For a *start-and-sell* entrepreneur, the Bellman system implies

$$V^s(0, \theta) = \frac{\beta\theta p(\bar{z})}{(1-\beta)(1+\beta\theta)}.$$

For a *start-and-manage* entrepreneur, letting π_m^* denote optimal per-period operating profit, the Bellman system implies

$$V^m(0, \theta) = \frac{\beta\theta \pi_m^*}{(1-\beta)(1-\beta\nu + \beta\theta)}.$$

Both $V^s(0, \theta)$ and $V^m(0, \theta)$ are strictly increasing in θ , so the relevant cutoffs are uniquely pinned down by pairwise indifference conditions.

Step 2: Occupational cutoffs. Comparing $V^s(0, \theta)$ and V^w yields the worker–seller cutoff:

$$\begin{aligned} V^s(0, \theta) > V^w &\iff \frac{\beta\theta p(\bar{z})}{(1-\beta)(1+\beta\theta)} > \frac{w}{1-\beta} \iff \frac{\beta\theta p(\bar{z})}{1+\beta\theta} > w \\ &\iff \beta\theta(p(\bar{z}) - w) > w \iff \theta > \theta^{sw} \equiv \frac{w}{\beta(p(\bar{z}) - w)}. \end{aligned}$$

Comparing $V^m(0, \theta)$ and V^w yields the worker–manager cutoff:

$$\begin{aligned} V^m(0, \theta) > V^w &\iff \frac{\beta\theta \pi_m^*}{(1-\beta)(1-\beta\nu + \beta\theta)} > \frac{w}{1-\beta} \iff \frac{\beta\theta \pi_m^*}{1-\beta\nu + \beta\theta} > w \\ &\iff \beta\theta(\pi_m^* - w) > (1-\beta\nu)w \iff \theta > \theta^{mw} \equiv \frac{(1-\beta\nu)w}{\beta(\pi_m^* - w)}. \end{aligned}$$

Finally, comparing $V^s(0, \theta)$ and $V^m(0, \theta)$ yields the seller–manager cutoff:

$$\begin{aligned} V^s(0, \theta) > V^m(0, \theta) &\iff \frac{p(\bar{z})}{1 + \beta\theta} > \frac{\pi_m^*}{1 - \beta\nu + \beta\theta} \\ &\iff \beta\theta(p(\bar{z}) - \pi_m^*) > \pi_m^* - (1 - \beta\nu)p(\bar{z}) \\ &\iff \theta > \theta^{sm} \equiv \frac{(1 - \beta\nu)p(\bar{z}) - \pi_m^*}{\beta(\pi_m^* - p(\bar{z}))}. \end{aligned}$$

Step 3: Sufficient conditions for all three occupations to be active. To obtain the ordering

$$\theta < \theta^{mw} \Rightarrow \text{worker}, \quad \theta^{mw} \leq \theta < \theta^{sm} \Rightarrow \text{manager}, \quad \theta \geq \theta^{sm} \Rightarrow \text{seller},$$

it suffices that (i) $\theta^{mw}, \theta^{sm} \in (0, 1)$ and (ii) $\theta^{mw} < \theta^{sm}$.

First, $\theta^{mw} \leq 1$ is equivalent to

$$\frac{(1 - \beta\nu)w}{\beta(\pi_m^* - w)} \leq 1 \iff w \leq \frac{\beta}{1 - \beta\nu + \beta} \pi_m^*,$$

and we also require $\pi_m^* > w$ so that the denominator is positive.

Second, to express θ^{sm} in terms of primitives, note that the bank's purchase price satisfies

$$p(\bar{z}) = \frac{\pi_b^*}{1 - \beta\nu}, \quad \pi_b^* = \lambda^{\frac{\mu}{1-\mu}} \pi_m^*,$$

so that $(1 - \beta\nu)p(\bar{z}) = \lambda^{\frac{\mu}{1-\mu}} \pi_m^*$. Substituting into θ^{sm} yields

$$\theta^{sm} = \frac{(1 - \beta\nu) \left(\lambda^{\frac{\mu}{1-\mu}} - 1 \right)}{\beta \left(1 - \beta\nu - \lambda^{\frac{\mu}{1-\mu}} \right)}.$$

Imposing $\theta^{sm} \leq 1$ is equivalent to the lower bound

$$\lambda^{\frac{\mu}{1-\mu}} \geq \frac{(1 + \beta)(1 - \beta\nu)}{1 - \beta\nu + \beta}.$$

Third, the existence of a nonempty manager region requires $\theta^{mw} < \theta^{sm}$, which is equivalent to

$$\theta^{mw} < \theta^{sm} \iff \beta\nu w < \pi_m^* \left(1 - \lambda^{\frac{\mu}{1-\mu}} \right) \iff \lambda^{\frac{\mu}{1-\mu}} < 1 - \frac{\beta\nu w}{\pi_m^*}.$$

(Under the same condition one can also verify $\theta^{mw} < \theta^{sw}$, so sellers do not appear below the worker–manager cutoff.)

Combining these inequalities delivers the sufficient conditions stated in Proposition 1. $\square$

B. Proof of Proposition 2.

Proof. Under a realization-based capital-gains tax at rate τ , the seller receives the after-tax sale price $(1 - \tau)p(\bar{z})$, so the seller–manager cutoff becomes

$$\theta_{\text{real}}^{sm} = \frac{(1 - \beta\nu)(1 - \tau)p(\bar{z}) - \pi_m^*}{\beta(\pi_m^* - (1 - \tau)p(\bar{z}))}.$$

Using $(1 - \beta\nu)p(\bar{z}) = \lambda^{\frac{\mu}{1-\mu}}\pi_m^*$, define $\kappa \equiv \lambda^{\frac{\mu}{1-\mu}}$ and rewrite

$$\theta_{\text{real}}^{sm} = \frac{(1 - \beta\nu)((1 - \tau)\kappa - 1)}{\beta((1 - \beta\nu) - (1 - \tau)\kappa)}, \quad \theta^{sm} = \frac{(1 - \beta\nu)(\kappa - 1)}{\beta((1 - \beta\nu) - \kappa)}.$$

A direct comparison yields

$$\theta_{\text{real}}^{sm} - \theta^{sm} = \frac{(1 - \beta\nu)\kappa\nu\tau}{((1 - \beta\nu) - (1 - \tau)\kappa)((1 - \beta\nu) - \kappa)}.$$

Under the maintained assumption that τ is small enough that all three occupations remain active, we have $\kappa > 1 - \beta\nu$ and $(1 - \tau)\kappa > 1 - \beta\nu$, implying

$$(1 - \beta\nu) - \kappa < 0 \quad \text{and} \quad (1 - \beta\nu) - (1 - \tau)\kappa < 0.$$

Hence the denominator is positive, while the numerator is strictly positive for $\tau > 0$. Therefore,

$$\theta_{\text{real}}^{sm} - \theta^{sm} > 0,$$

which proves Proposition 2. $\square$

C. Proof of Proposition 3.

Proof. As in the main text, let $\kappa \equiv \lambda^{\frac{\mu}{1-\mu}}$ and use $(1 - \beta\nu)p(\bar{z}) = \kappa\pi_m^*$. Under accrual taxation at

rate τ , the seller–manager cutoff can be written as

$$\theta_{\text{accrual}}^{sm} = \frac{\kappa((1 - \beta\nu) - \beta\tau(1 - \nu)) - (1 - \beta\nu)}{\beta((1 - \beta\nu) - (1 - \beta\tau)\kappa)}.$$

Step 1: $\theta_{\text{accrual}}^{sm} > \theta^{sm}$. A direct comparison yields

$$\theta_{\text{accrual}}^{sm} - \theta^{sm} = \frac{\nu \tau \kappa [(1 - \beta\nu) - (1 - \beta)\kappa]}{((1 - \beta\nu) - (1 - \beta\tau)\kappa)((1 - \beta\nu) - \kappa)}.$$

Under the parameter restrictions for the three-occupation case, $\kappa > 1 - \beta\nu$, so both denominator terms are negative for sufficiently small τ , and their product is positive. The numerator is positive for $\tau > 0$ under the same restrictions. Hence

$$\theta_{\text{accrual}}^{sm} - \theta^{sm} > 0.$$

Step 2: $\theta_{\text{accrual}}^{sm} < \theta_{\text{real}}^{sm}$. Comparing the accrual and realization cutoffs yields

$$\theta_{\text{accrual}}^{sm} - \theta_{\text{real}}^{sm} = -\frac{\nu \tau (1 - \tau) (1 - \beta) \kappa^2}{((1 - \beta\nu) - (1 - \beta\tau)\kappa)((1 - \beta\nu) - (1 - \tau)\kappa)} < 0,$$

again using that the denominator is positive for sufficiently small τ in the region where all three occupations remain active.

Combining Steps 1 and 2 delivers

$$\theta_{\text{real}}^{sm} > \theta_{\text{accrual}}^{sm} > \theta^{sm}.$$

Step 3: $\theta_{\text{accrual}}^{mw} > \theta^{mw}$. In the main text,

$$\theta^{mw} = \frac{(1 - \beta\nu)w}{\beta(\pi_m^* - w)}, \quad \theta_{\text{accrual}}^{mw} = \frac{(1 - \beta\nu)w}{\beta(\pi_m^* - w - (1 - \beta)\tau p(\bar{z}))}.$$

Since $(1 - \beta)\tau p(\bar{z}) > 0$, the denominator of $\theta_{\text{accrual}}^{mw}$ is strictly smaller than that of θ^{mw} (and is

positive for sufficiently small τ), implying

$$\theta_{\text{accrual}}^{mw} > \theta^{mw}.$$

Finally, under realization-based taxation the worker–manager cutoff is unchanged, so $\theta_{\text{real}}^{mw} = \theta^{mw}$, completing the proof. $\square$